

WEEKLY REPORT 5

GROUP MEMBER AND ROLL NO

- Yash Kirti AU2340149
- Yashvi Modi AU2340116
- Priyanshu Rupala AU2340207
- Tathya Parikh AU2340151

Week: 2 (Super-Resolution Implementation Phase) Project Title: Super Resolution using Classical Interpolation Methods

Objective

This week was dedicated to the quantitative measures of the complete classical SR pipeline as well as qualitative visual comparison of its performance. It was hoped to determine the ability of each method to reconstruction of edges, textures, and terrain structures in satellite images.

Work Done

a. Quantitative Evaluation

On all 10 satellite images, PSNR and SSIM were determined on four SR methods.
Method

Method	Avg PSNR (dB)	Avg SSIM	Observations
Nearest	28.4	0.44	Blocky artifacts, poor fidelity
Bilinear	29.6	0.50	Smooth transitions, blurred textures
Bicubic	31.0	0.56	Balanced results but loss of high-frequency data
NEDI + IBP + Wavelet	33.9	0.69	Clear textures, sharp edges, visually closest to HR

The effectiveness of the combined approach is proven by 2-3 dB PSNR improvements and by an approximate of 0.13 growth in SSIM.

All the measurements were calculated based on the channel-wise comparisons and Scikit-image library functions.

b. Visual Analytical Quality

Editing and interpreting the photographs.

Drawn comparative panels of each type of terrain:

Urban: Curb and structure were brought to visibility.

Forest: Patterns of leaves and canopies were smoother and had less artifacts.

Desert: Dusts of fine sand texture worked well without drying out with excess enhancement effect.

Water: Grade flat, rhymed with slight ringing.

Developed multi-pane visualization (HR, LR, Bicubic SR, Final SR) to add to the final presentation.

c. Documentation & Reporting

Entered all the results in a tabulated format in the report.

Graph of PSNR and SSIM provided by Matplotlib bar.

Wrote the drafts of "Results" and "Conclusion" sections to be submitted finally.

Observations

The hybrid NEDI + IBP + Wavelet method always left all interpolation baselines far behind.

Results Rate variation with terrain showed that the technique can be customized to a wide variety of feature densities.

Bicubic interpolation was still the fastest in computation (although less pleasing to the eye).

The average increase in the new method was 9%-12% in perceptual.

Challenges

Minor color distortion observed on the super-saturated satellite images (as a result of wavelet coefficient scaling).

Large memory consumption when IBP iterating great data sets.

Next Set Work

Use subject-specific post-enhancement (denoising + contrast equalization).

Write final report with all graphs, table and figures.

Compare methods on a run time basis.

Produce final presentations slides on official format of the AU.

Close by limitations and scope of future.